

Program:

```
import nltk
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
# Function to perform basic text processing operations
def text_processing(text):
    # Tokenization
    tokens = word_tokenize(text)

    # Sentence segmentation
    sentences = sent_tokenize(text)

    # Stopword removal
    stop_words = set(stopwords.words("english"))
    filtered_tokens = [token for token in tokens if token.lower() not in stop_words]

    # Stemming
    stemmer = PorterStemmer()
    stemmed_tokens = [stemmer.stem(token) for token in filtered_tokens]

    # Lemmatization
    lemmatizer = WordNetLemmatizer()
    lemmatized_tokens = [lemmatizer.lemmatize(token) for token in filtered_tokens]

    return tokens, sentences, filtered_tokens, stemmed_tokens, lemmatized_tokens

# Example usage
text = "This is an example sentence. It demonstrates basic text processing operations."
```

```
tokens, sentences, filtered_tokens, stemmed_tokens, lemmatized_tokens =  
text_processing(text)
```

```
print("Original Text:")  
print(text)  
print("")
```

```
print("Tokens:")  
print(tokens)  
print("")
```

```
print("Sentences:")  
print(sentences)  
print("")
```

```
print("Filtered Tokens (after stopword removal):")  
print(filtered_tokens)  
print("")
```

```
print("Stemmed Tokens:")  
print(stemmed_tokens)  
print("")
```

```
print("Lemmatized Tokens:")  
print(lemmatized_tokens)
```

Output:

Original Text:

This is an example sentence. It demonstrates basic text processing operations.

Tokens:

['This', 'is', 'an', 'example', 'sentence', '.', 'It', 'demonstrates', 'basic', 'text', 'processing', 'operations', '.']

Sentences:

['This is an example sentence.', 'It demonstrates basic text processing operations.']

Filtered Tokens (after stopword removal):

['This', 'example', 'sentence', '.', 'It', 'demonstrates', 'basic', 'text', 'processing', 'operations', '.']

Stemmed Tokens:

['thi', 'exampl', 'sentenc', '.', 'It', 'demonstr', 'basic', 'text', 'process', 'oper', '.']

Lemmatized Tokens:

['This', 'example', 'sentence', '.', 'It', 'demonstrates', 'basic', 'text', 'processing', 'operation', '.']